

Geometry

Name\_\_\_\_\_

Unit 1

Date\_\_\_\_\_

Lesson 10 Homework

Period\_\_\_\_\_

$\overline{LM}$  bisects  $\overline{ZT}$  at point S.  $\overline{ZS}$  has a length of  $3x + 7$ , and  $\overline{ST}$  has a length of  $-5x + 23$ .

1. Draw a diagram of the description.

2. Find the value of  $x$ .

3. Determine the length of  $\overline{ZT}$ .

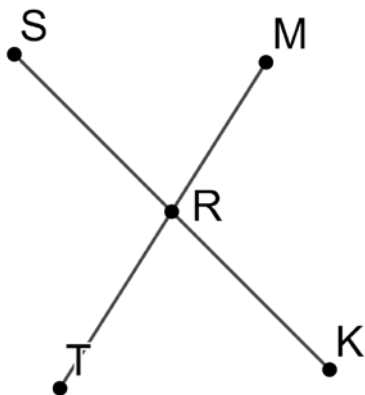
$\overline{XY}$  bisects  $\overline{WB}$  at point  $L$ .  $WB = 3x + 8$  and  $WL = \frac{1}{2}x + 12$ .

4. Draw a diagram of the description.

5. Find the value of  $x$ .

6. Find the length of  $\overline{WB}$ .

$\overline{MT}$  bisects  $\overline{SK}$  at point  $R$  as shown.  $\overline{SK}$  has a length of  $24x - 12$ ,  $\overline{RK}$  has a length of  $10x - 2$ ,  $\overline{TR}$  has a length of  $4x + 5$ , and  $\overline{RM}$  has a length of  $8x - 5$ .



7. Based on the given information which two segments are congruent?
8. Find the value of  $x$ .
9. Find the length of  $\overline{RK}$ .
10. Find the length of  $\overline{MT}$ .